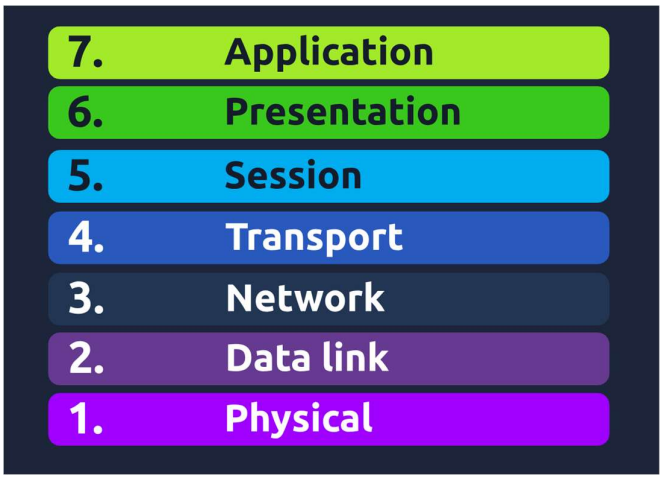
Room – Networking Concepts

OSI Layer



Layer 1: Physical Layer

The physical layer, also referred to as layer 1, deals with the physical connection between devices; this includes the medium, such as a wire, and the definition of the binary digits 0 and 1. Data transmission can be via an electrical, optical, or wireless signal. Consequently, we need data cables or antennas, depending on our physical medium.

Layer 2: Data Link Layer

The Data Link Layer (Layer 2) in the OSI model is responsible for node-to-node data transfer, ensuring error-free transmission between adjacent network nodes. It converts packets from the network layer into frames, manages physical addressing (MAC addresses)

Layer 3: Network Layer

layer 3, is concerned with sending data between different networks. In more technical terms, the network layer handles logical addressing and routing, i.e., finding a path to transfer the network packets between the diverse networks.

Layer 4: Transport Layer

Layer 4, the transport layer, enables end-to-end communication between running applications on different hosts. Your web browser is connected to the TryHackMe web server over the transport layer, which can support various functions like flow control, segmentation, and error correction.

Examples of layer 4 are Transmission Control Protocol (TCP) and User Datagram Protocol (UDP).

Layer 5: Session Layer

The session layer is responsible for establishing, maintaining, and synchronising communication between applications running on different hosts. Establishing a session means initiating communication between applications and negotiating the necessary parameters for the session

Layer 6: Presentation Layer

The presentation layer ensures the data is delivered in a form the application layer can understand. Layer 6 handles data encoding, compression, and encryption. An example of encoding is character encoding, such as ASCII or Unicode.

Layer 7: Application Layer

The application layer provides network services directly to end-user applications. Your web browser would use the HTTP protocol to request a file, submit a form, or upload a file.

TCP/IP Model

- **Application Layer**: The OSI model application, presentation and session layers, i.e., layers 5, 6, and 7, are grouped into the application layer in the TCP/IP model.
- **Transport Layer**: This is layer 4.
- **Internet Layer**: This is layer 3. The OSI model's network layer is called the Internet layer in the TCP/IP model.
- **Link Layer**: This is layer 2.

Layer Number	ISO OSI Model	TCP/IP Model (RFC 1122)
7	Application Layer	Application Layer
6	Presentation Layer	
5	Session Layer	
4	Transport Layer	Transport Layer
3	Network Layer	Internet Layer
2	Data Link Layer	Link Layer
1	Physical Layer	

Real world systems use this .

What is an IP Address?

An **IPv4 address**:

- Has **32 bits**
- Written as **4 octets**
- Each octet = number between **0–255**
- Example: 192.168.0.1

Total possible IPv4 addresses ≈

$$2^{32} \approx 4.3 \text{ billion}$$

Every device on a network needs a **unique IP address** so other devices can find it.

Network & Broadcast Addresses

In a subnet like:

192.168.1.0/24

- 192.168.1.0 → Network address
- 192.168.1.255 → Broadcast address
- Usable hosts → 192.168.1.1 to 192.168.1.254

You can't assign .0 or .255 to devices.
They're reserved.

Public vs Private IP

Public IP

- Used on the Internet
- Globally unique

Private IP (RFC 1918 ranges)

Range	CIDR
10.0.0.0 – 10.255.255.255	10/8
172.16.0.0 – 172.31.255.255	172.16/12
192.168.0.0 – 192.168.255.255	192.168/16

Private IPs **cannot directly access the Internet**.
They need a router with **NAT** to translate them.

Subnetting is just:

Dividing a big network into smaller networks.

What Does /24 Mean?

When you see:

192.168.66.89/24

The /24 means:

- First **24 bits are network part**
- Remaining **8 bits are host part**

Since IPv4 is 32 bits:

24 bits (network) + 8 bits (host) = 32 bits

8 host bits means:

$$2^8 = 256 \text{ total addresses}$$

But:

- 1 is network address
- 1 is broadcast address

So usable hosts:

$$256 - 2 = 254$$

That's why in /24, usable IPs = 254.

Router's Role

A router:

- Looks at IP address
- Forwards packet to correct network
- Works at **Layer 3 (Network Layer)**

Packets may pass through multiple routers before reaching destination.

The IP protocol allows us to reach a destination host on the network; the host is identified by its IP address. We need protocols that would enable processes on networked hosts to communicate with each other. There are two transport protocols to achieve that: UDP and TCP.

UDP

UDP (User Datagram Protocol) allows us to reach a specific process on this target host. UDP is a simple connectionless protocol that operates at the transport layer, i.e., layer 4. Being connectionless means that it does not need to establish a connection. UDP does not even provide a mechanism to know that the packet has been delivered.

TCP

TCP (Transmission Control Protocol) is a connection-oriented transport protocol. It uses various mechanisms to ensure reliable data delivery sent by the different processes on the networked hosts. Like UDP, it is a layer 4 protocol. Being connection-oriented, it requires the establishment of a TCP connection before any data can be sent.

A TCP connection is established using what's called a three-way handshake. Two flags are used: SYN (Synchronise) and ACK (Acknowledgment). The packets are sent as follows:

1. SYN Packet: The client initiates the connection by sending a SYN packet to the server. This packet contains the client's randomly chosen initial sequence number.
2. SYN-ACK Packet: The server responds to the SYN packet with a SYN-ACK packet, which adds the initial sequence number randomly chosen by the server.

3. ACK Packet: The three-way handshake is completed as the client sends an ACK packet to acknowledge the reception of the SYN-ACK packet.

The Life of a Packet

Based on what we have studied so far, we can explain a *simplified version* of the packet's life. Let's consider the scenario where you search for a room on TryHackMe.

1. On the TryHackMe search page, you enter your search query and hit enter.
2. Your web browser, using HTTPS, prepares an HTTP request and pushes it to the layer below it, the transport layer.
3. The TCP layer needs to establish a connection via a three-way handshake between your browser and the TryHackMe web server. After establishing the TCP connection, it can send the HTTP request containing the search query. Each TCP segment created is sent to the layer below it, the Internet layer.
4. The IP layer adds the source IP address, i.e., your computer, and the destination IP address, i.e., the IP address of the TryHackMe web server. For this packet to reach the router, your laptop delivers it to the layer below it, the link layer.
5. Depending on the protocol, The link layer adds the proper link layer header and trailer, and the packet is sent to the router.
6. The router removes the link layer header and trailer, inspects the IP destination, among other fields, and routes the packet to the proper link. Each router repeats this process until it reaches the router of the target server.

On a WiFi, within what will an IP packet be encapsulated?

Frame

What do you call the UDP data unit that encapsulates the application data?

Datagram

What do you call the data unit that encapsulates the application data sent over TCP?

Segment

The TELNET (Teletype Network) protocol is a network protocol for remote terminal connection. In simpler words, telnet, a TELNET client, allows you to connect to and communicate with a remote system and issue text commands. Although initially it was used for remote administration, we can use telnet to connect to any server listening on a TCP port number.